

RobustGaSP: MLE vs MAP estimation

$$y = \sqrt{2 - x_1^2 - x_2^2} \text{ vs } y = \sqrt{2 - (x_1 - 1)^2 - x_2^2}$$

2D test example

1 Function 1: $y = \sqrt{2 - x_1^2 - x_2^2}$

1.1 Emulator fit and validation

We have a 2-dimensional input, one output, and 20 noise-free observations (Figure 1).

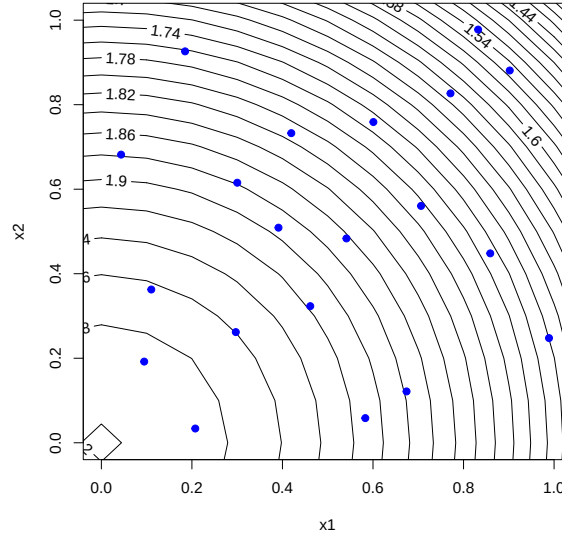


Figure 1: Contour plot for the function $y = \sqrt{2 - x_1^2 - x_2^2}$, with the location in the 2D input space of the observations' input settings shown in blue.

Going forward, we will assume a linear mean function of the form $m(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. The Gauss (squared exponential) *and* Matern 5/2 kernels will be compared for both estimation methods.

100 new LH points are generated, and out-of-sample checks (as per **RobustGaSP's** Help page) shown in Table 1, with the emulator predictions plotted against the actual outputs for these points shown in Figure 2.

Table 1: Out-of-sample checks

Estimate method	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
MLE	Gauss	0.57	0.00006	0.0117
MLE	Matern 5/2	0.86	0.00007	0.0117
MAP	Gauss	0.4	0.00007	0.0086
MAP	Matern 5/2	0.86	0.00006	0.0112

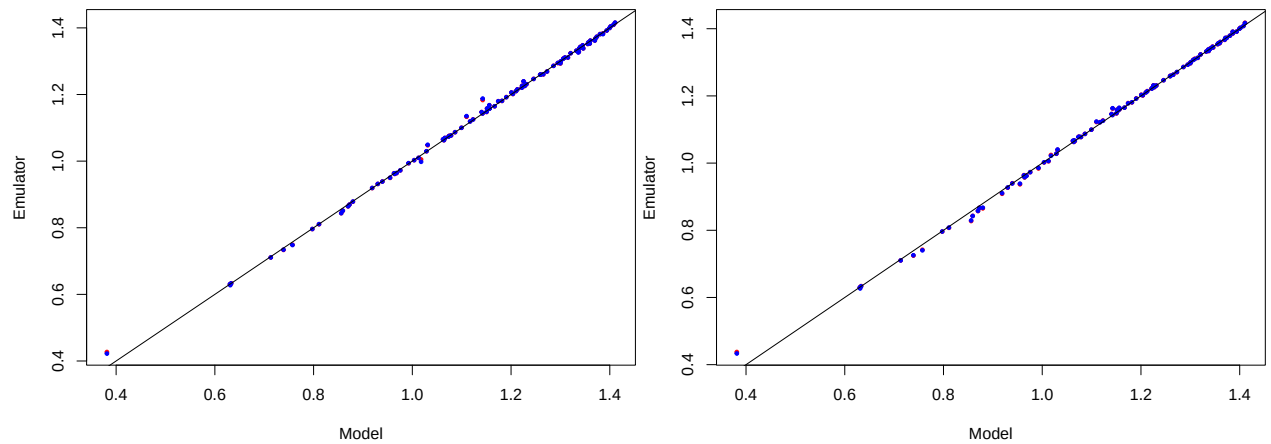


Figure 2: Emulator versus model for the new points. Left: Using a Gauss kernel. Right: Using a Matern 5/2 kernel. Red: MLE. Blue: MAP

Finally, in preparation for the real world data where we cannot perform out-of-sample validation, leave-one-out validation is carried out; the results are shown in Figure 3.

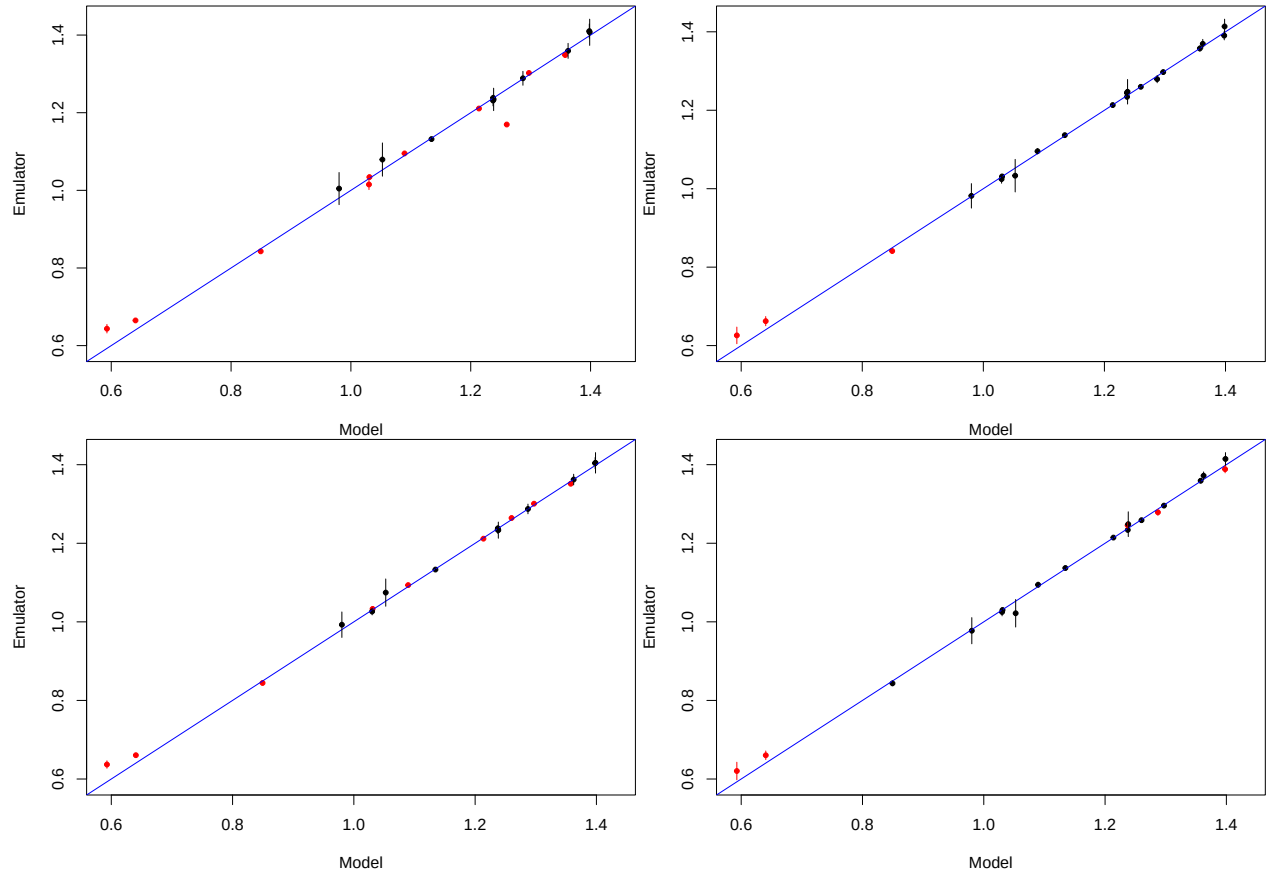


Figure 3: LOO cross validation. Top: Maximum likelihood estimation for the Gauss (left) and Matern 5/2 (right) kernels. Bottom: Marginal posterior mode estimation.

Going forward, for now, only the Gauss kernel is dealt with; Matern 5/2 will be returned to once I've figured out the estimates returned by the two packages and the derivative of the Matern 5/2 kernel. The estimates returned by both estimation methods using the Gauss kernel are shown in Table 2.

Table 2: Estimates table

Estimation	Kernel	Trend 1	Trend 2	Trend 3	Range 1	Range 2	Variance
MLE	Gauss	1.54	-0.7	-1.01	0.81	0.63	0.09
MAP	Gauss	1.45	-0.94	-1.95	1.18	0.84	0.82

1.2 Verifying the estimates by-hand for both estimation methods

To calculate the posterior predictions at the 100 testing points by-hand, we require the matrix H containing the inputs for the 20 observations and well as matrix A , its inverse, and $\mathbf{t}(x^*)$ for all 1000 of the new inputs x^* . A and $\mathbf{t}(x^*)$ are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by the two estimation methods (Figure 4).

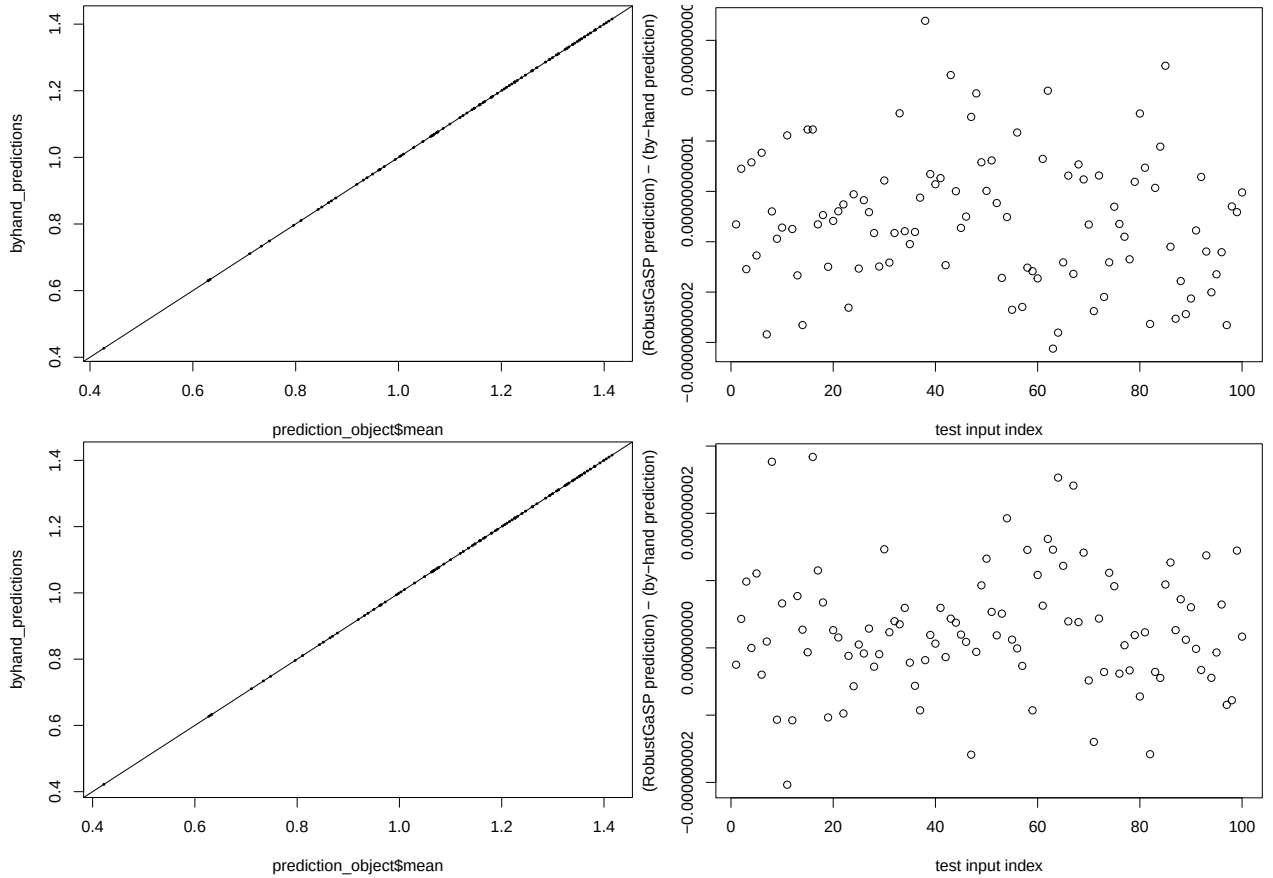


Figure 4: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE. Bottom: MAP.

For both we calculate

$$\frac{\sum_{i=1}^{1000} \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{1000}$$

then divide this by 10^{-8} (machine precision) to get 0.001 for MLE and 0.07 for MAP.

We create an object containing the emulated partial derivatives for use in the alignment measure, and check that these approximately match the partial derivatives of the actual function (Figure 5). Also note that the two estimation methods return similar estimates for the partial derivatives (Figure 6).

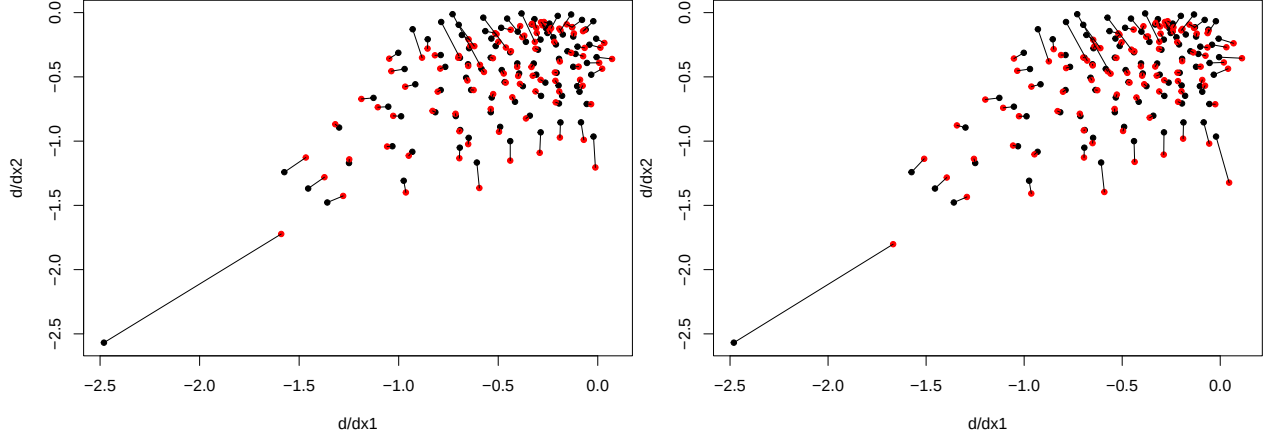


Figure 5: The estimated partial derivatives found by each estimation method (red dots, left MLE and right MAP) match the true partial derivatives (black) well.

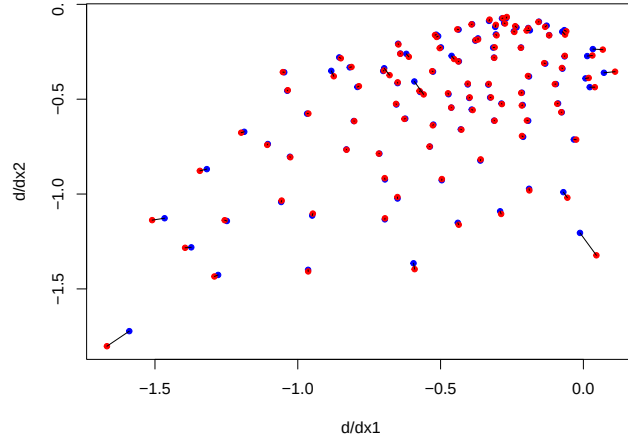


Figure 6: The estimated partial derivatives returned by each estimation method are similar (blue for MLE and red for MAP).

Finally, we create an object containing the normalised partial derivatives for use in the alignment measure.

2 Function 2: $y = \sqrt{2 - (x_1 - 1)^2 - x_2^2}$

We repeat the above steps for this new function (Figure 7). The same training and testing points are used as for the first function.

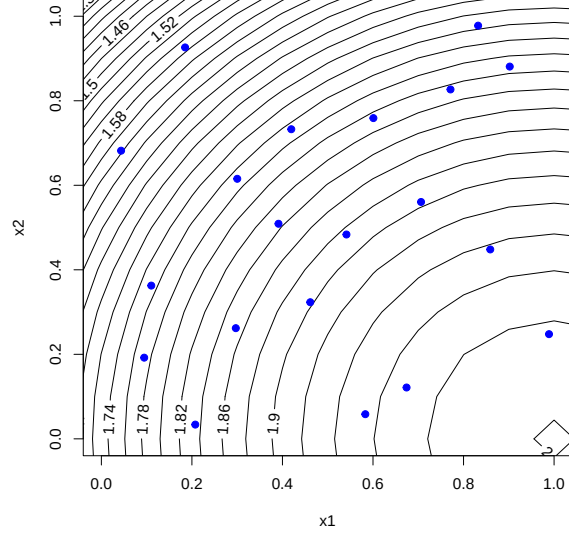


Figure 7: Contour plot for the function $y = \sqrt{2 - (x_1 - 1)^2 - x_2^2}$, with the location in the 2D input space of the observations' input settings shown in blue.

2.1 Emulator fit and validation

Out-of-sample check results are shown in Table 3 and Figure 8.

Table 3: Out-of-sample checks

Estimate method	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
Maximum likelihood	Gauss	0.68	0.0002	0.003
Maximum likelihood	Matern 5/2	0.64	0.0002	0.0062
MAP	Gauss	0.6	0.0001	0.0026
MAP	Matern 5/2	0.58	0.0002	0.0056

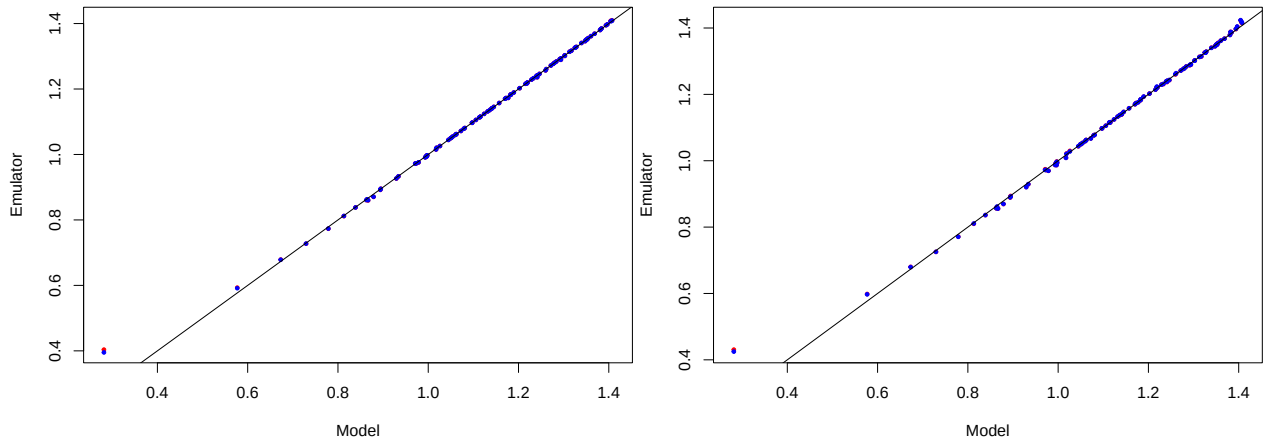


Figure 8: Emulator versus model for the new points. Left: Using a Gauss kernel. Right: Using a Matern 5/2 kernel. Red: MLE. Blue: MAP

LOO CV results are shown in Figure 9.

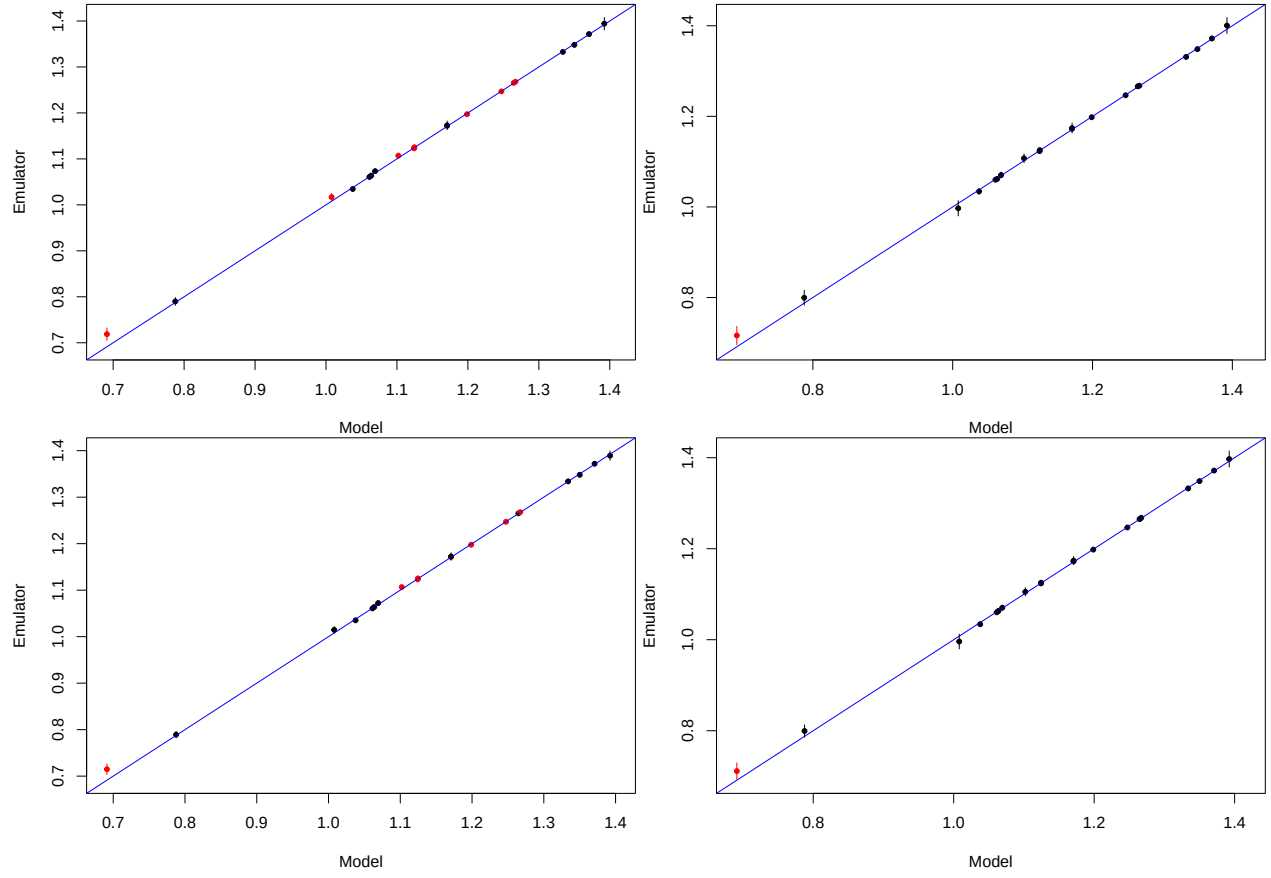


Figure 9: LOO cross validation. Top: Maximum likelihood estimation for the Gauss (left) and Matern 5/2 (right) kernels. Bottom: Marginal posterior mode estimation for the Gauss (left) and Matern 5/2 (right) kernels.

Parameter estimates using the Gauss kernel are shown in Table 4.

Table 4: Estimates table

Estimation	Kernel	Trend 1	Trend 2	Trend 3	Range 1	Range 2	Variance
MLE	Gauss	0.4	0.81	-1.03	1.11	0.96	0.2
MAP	Gauss	-0.23	1.18	-1.58	1.43	1.19	1.06

2.2 Verifying the estimates by-hand for both estimation methods

A good match is found for both the predictions (Figure 10) and the partial derivatives (Figure 11, top row). The two estimation methods produce similar estimates for the partial derivatives (Figure 11, bottom row).

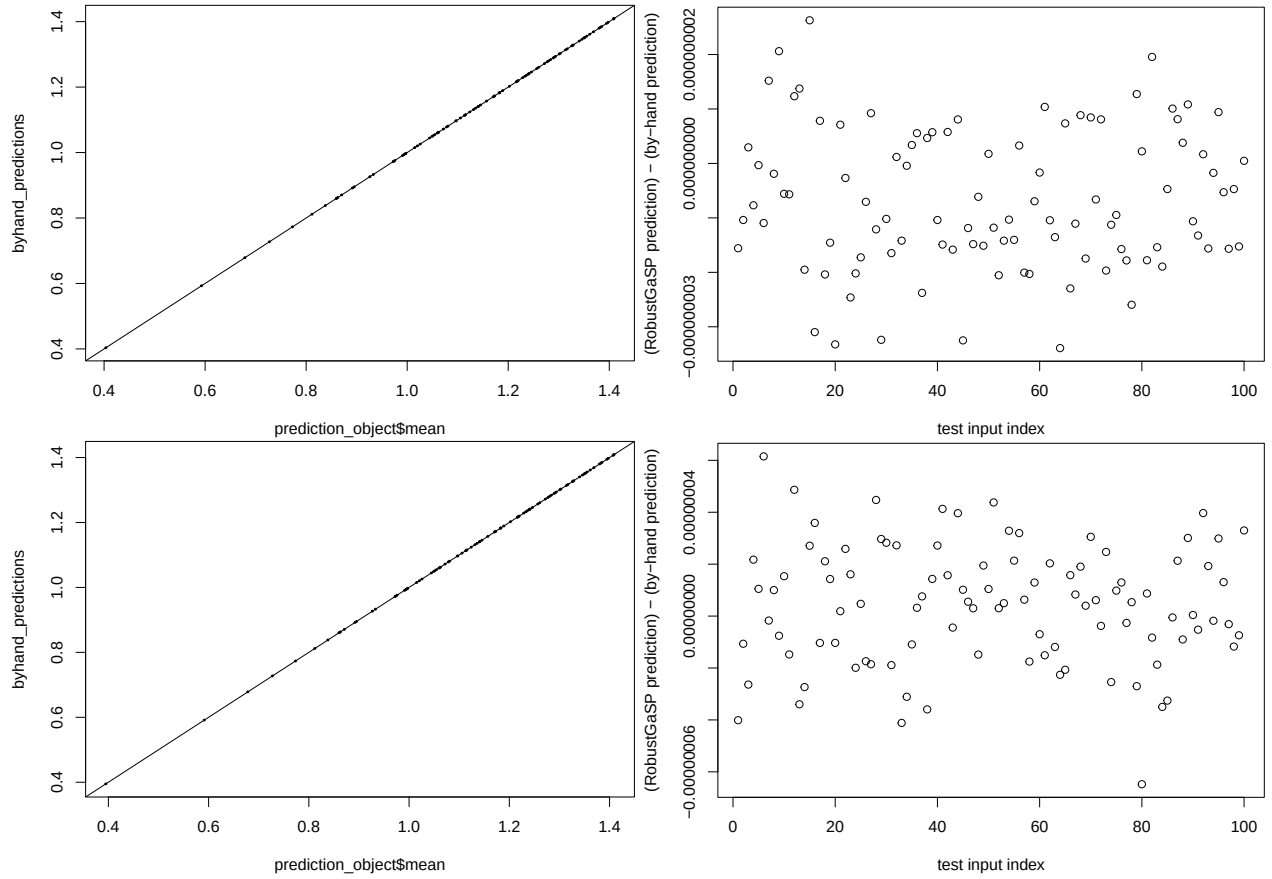


Figure 10: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE. Bottom: MAP.

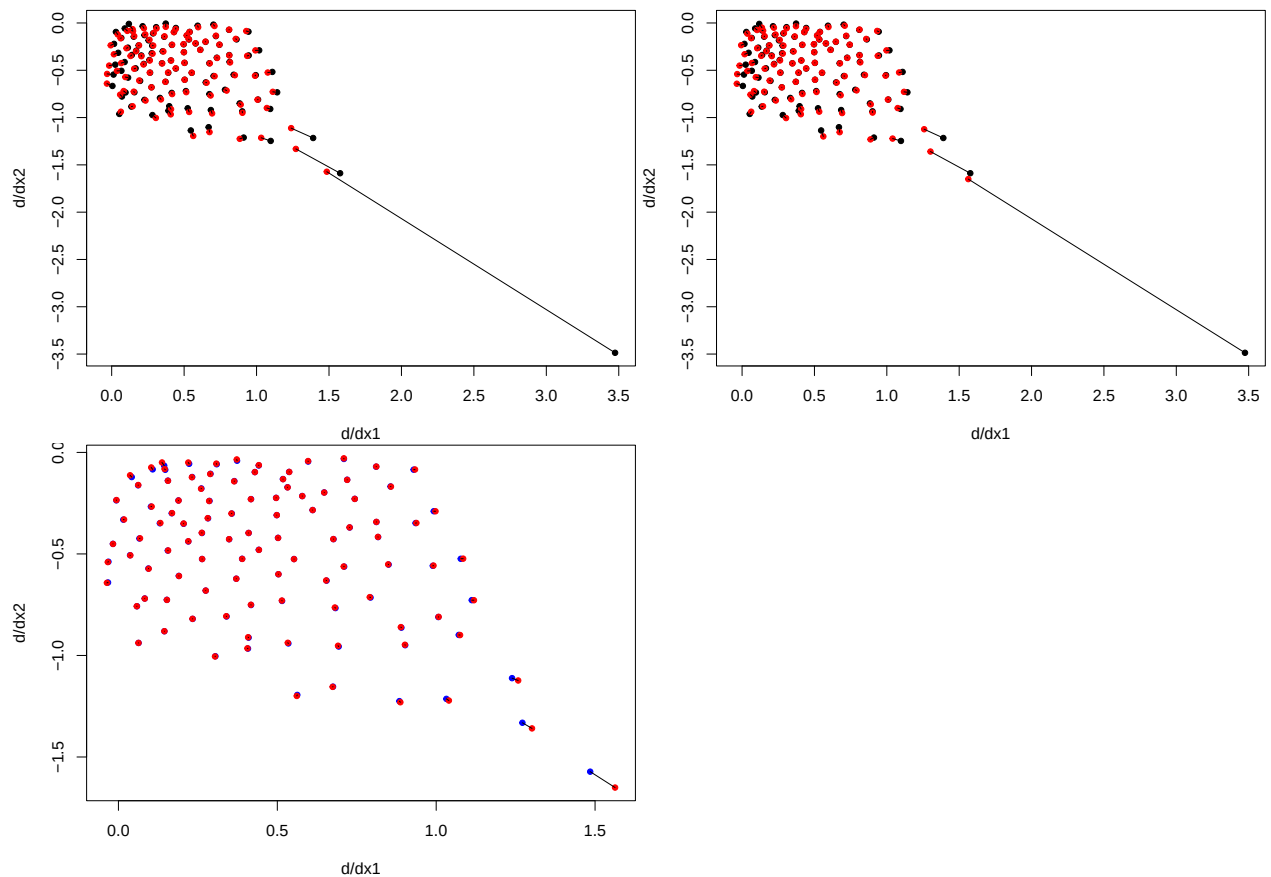


Figure 11: Top row: The estimated partial derivatives found by each estimation method (red dots, left MLE and right MAP) match the true partial derivatives (black) well. Bottom plot: The estimated partial derivatives returned by each estimation method are similar (blue for MLE and red for MAP).

3 Alignment measure

We calculate the alignment measure of this function with a couple of others. As can be seen in Figure 13, where the two functions are superimposed, there is a large amount of the 2D input space in which the response surfaces are more orthogonal than they are parallel.

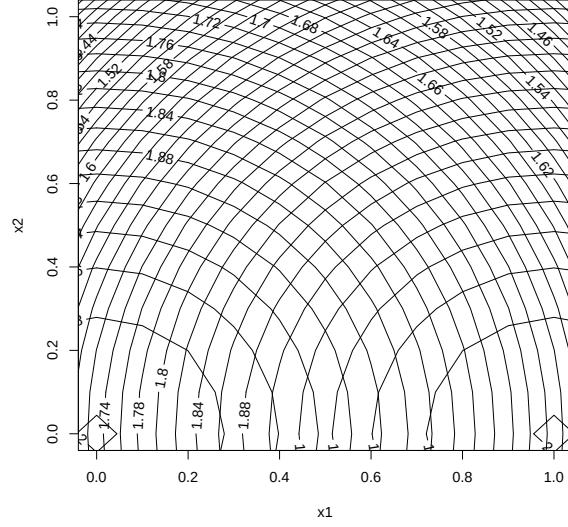


Figure 12: Contour plots for the functions $y = \sqrt{2 - x_1^2 - x_2^2}$ and $f(x_1, x_2) = \sqrt{2 - (x_1 - 1)^2 - x_2^2}$.

Using the **actual** partial derivatives of the two functions at the 100 testing point, we find an AM of 0.4493 (and 0.4486 using 1000). Using MLE at these 100 points we get an alignment for these two functions of 0.4405, (and 0.4387 using 1000) and for MAP 0.4444, (and 0.4416 using 1000).